Code of the student

☐0 ☐0 ☐0 ☐0 ☐0 ☐0 ☐0 ☐0
☐1 ☐1 ☐1 ☐1 ☐1 ☐1 ☐1 ☐1
☐2 ☐2 ☐2 ☐2 ☐2 ☐2 ☐2 ☐2
☐3 ☐3 ☐3 ☐3 ☐3 ☐3 ☐3 ☐3
☐4 ☐4 ☐4 ☐4 ☐4 ☐4 ☐4 ☐4
☐5 ☐5 ☐5 ☐5 ☐5 ☐5 ☐5 ☐5
☐6 ☐6 ☐6 ☐6 ☐6 ☐6 ☐6 ☐6
☐7 ☐7 ☐7 ☐7 ☐7 ☐7 ☐7 ☐7
☐8 ☐8 ☐8 ☐8 ☐8 ☐8 ☐8 ☐8
☐9 ☐9 ☐9 ☐9 ☐9 ☐9 ☐9 ☐9

Surname, Name:

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IP TUNNEL DIAGNOSIS QUIZ FOR OVNET

IMPORTANT REMARKS:

- It is important to fill the ID number of the student and the code of the test.
- IT IS COMPLETELY FORBIDDEN TO USE ANY KIND OF COMMUNICATIONS DEVICE.

```
1) #####  ConfTunnelAAA
2) # THESE ARE THE SAME COMMANDS THAN IN THE LAB SESSION
3) # COMMANDS EXECUTED IN R1
4) R1# iptables -t nat -A POSTROUTING -o eth2 \\
5) -j SNAT --to 192.0.2.2
6)
7) ## with label "addtun_nopmtu"
8) R1# ip tunnel add tunnel0 mode ipip local 192.0.2.2 \\
9) remote 198.51.100.2 ttl 0 nopmtudisc  dev eth2
10) R1# ifconfig tunnel0 0.0.0.0
11) R1# route add -net 172.16.1.0/24 dev tunnel0
12)
13) ## with label "addtun_pmtu"
14)
15) R1# ip tunnel add tunnel0 mode ipip local 192.0.2.2 \\
16) remote 198.51.100.2 ttl 0 pmtudisc  dev eth2
17) R1# ifconfig tunnel0 0.0.0.0
18) R1# route add -net 172.16.1.0/24 dev tunnel0
19)
20)
21) # COMMANDS EXECUTED IN R2
22) R2# iptables -t nat -A POSTROUTING -o eth2 \\
23) -j SNAT --to 198.51.100.2
24)
25) ## with label "addtun_nopmtu"
26) R2# ip tunnel add tunnel0 mode ipip local 198.51.100.2 \\
27) remote 192.0.2.2 ttl 0 nopmtudisc dev eth2
28) R2# ifconfig tunnel0 0.0.0.0
29) R2# route add -net 192.168.0.0/24 dev tunnel0
30)
31) ## with label "addtun_pmtu"
32) R2# ip tunnel add tunnel0 mode ipip local 198.51.100.2 \\
33) remote 192.0.2.2 ttl 0 pmtudisc dev eth2
34) R2# ifconfig tunnel0 0.0.0.0
35) R2# route add -net 192.168.0.0/24 dev tunnel0
```

Pregunta 1 Regarding ICMP relaying, as stated in RFC 2003, it is FALSE that:

☐ The action taken by the encapsulator does not depend on the type of ICMP message received, as it always will send an ICMP Destination Unreachable message towards the source

☐ The encapsulator may receive an ICMP message from any intermediate router within the tunnel other than the tunnel exit point

☐ When the received message contains enough information, the encapsulator MAY use the incoming ICMP message to create a similar ICMP message to be sent to the original sender.

☐ ICMP messages indicating an error in processing a datagram include a copy of (a portion of) the datagram causing the error.

Pregunta 2 In the scenario of Figure 1, we have introduced the configuration described in **ConfTunnelAAA**, which corresponds exactly to the configuration of the ip-tunnel lab session. The MTU of all the networks is 1500 Bytes, except **SimNet2** that is 996 Bytes. You execute the "addtun_nopmtu" label to create the tunnel with the `nopmtudisc` policy. You have executed the following filtering command in **R1**:
`iptables -A INPUT -p icmp -j DROP`
and now you send the file `/etc/services` from **host2** (behaving as TCP client) towards **host3** (behaving as TCP server) using `nc`. Then it is FALSE that:

☐ The TCP handshake segments (SYN, SYN/ACK, ACK) have no problems to be transmitted

☐ There are no Fragmentation Needed messages in **SimNet0**

☐ The TCP connection remains stalled until it is closed due to the number of retransmissions allowed is exceeded

☐ There are Fragmentation Needed messages in **SimNet1** to inform **R1** that the maximum MTU is 996

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Pregunta 3 In the scenario of Figure 1, we have introduced the configuration described in **ConfTunnelAAA**, which corresponds exactly to the configuration of the ip-tunnel lab session. The MTU of all the networks is 1500 Bytes, except **SimNet2** that is 996 Bytes. You execute the "addtun_pmtu" label to create the tunnel with the `pmtudisc` policy. If you execute the following command in **host2**:
`ping -c3 -s 1000 -M do -i1 172.16.1.3`
then it is FALSE that:

- ☐ There is a Fragmentation Needed message in **SimNet1** that notifies a maximum MTU of 996
- ☐ There is a Fragmentation Needed message in **SimNet0** that notifies a maximum MTU of 976
- ☐ The third ICMP Request message in **SimNet0** is sent fragmented in origin in two frames
- ☐ The two first ICMP Echo Request packets are sent with the DF flag Set

Pregunta 4 In the scenario of Figure 1, we have introduced the configuration described in **ConfTunnelAAA**, which corresponds exactly to the configuration of the ip-tunnel lab session. The MTU of all the networks is 1500 Bytes, except **SimNet2** that is 996 Bytes. You execute the "addtun_nopmtu" label to create the tunnel with the `nopmtudisc` policy. If you execute the following command in **host2**:
`ping -c3 -s 1000 -M want 172.16.1.3`
then it is FALSE that:

- ☐ No error is generated by the ICMP Echo Reply message
- ☐ Only one ICMP Echo Reply message is received in **host2**
- ☐ The ICMP Echo Reply message is fragmented in path
- ☐ The third ICMP Echo Request message in **SimNet0** is fragmented in origin, but the DF flag of these fragments is Set

Pregunta 5 In the scenario of Figure 1, we have introduced the configuration described in **ConfTunnelAAA**, which corresponds exactly to the configuration of the ip-tunnel lab session. You execute the "addtun_nopmtu" label to create the tunnel with the `nopmtudisc` policy. If you execute the following command in **host2**:
`host2$ ping -c1 -t3 172.16.1.3`
then it is FALSE that:

- ☐ In the ICMP Echo Request message in **SimNet2**, the value of the TTL in the Inner Header is 2
- ☐ The ping packet reaches the destination host
- ☐ In the ICMP Echo Request message in **SimNet1**, the value of the TTL in the Outer Header is 1
- ☐ In the ICMP Echo Reply message in **SimNet0**, the destination address corresponds to **host2**

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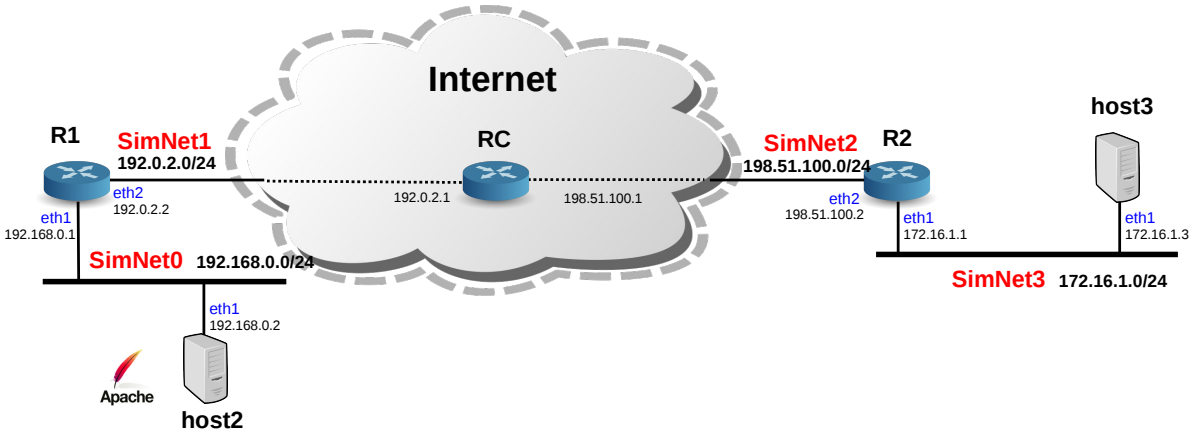


Fig. 1: IP tunnel scenario

Para revisión, preferentemente imprimir documentos compilados desde auto-multiple-choice.